

5. Time-Shifting Property

If $\mathcal{F}[f(t)] = F(\omega)$, Then $\mathcal{F}[f(t - t_0)] = F(\omega) e^{-j\omega t_0}$;

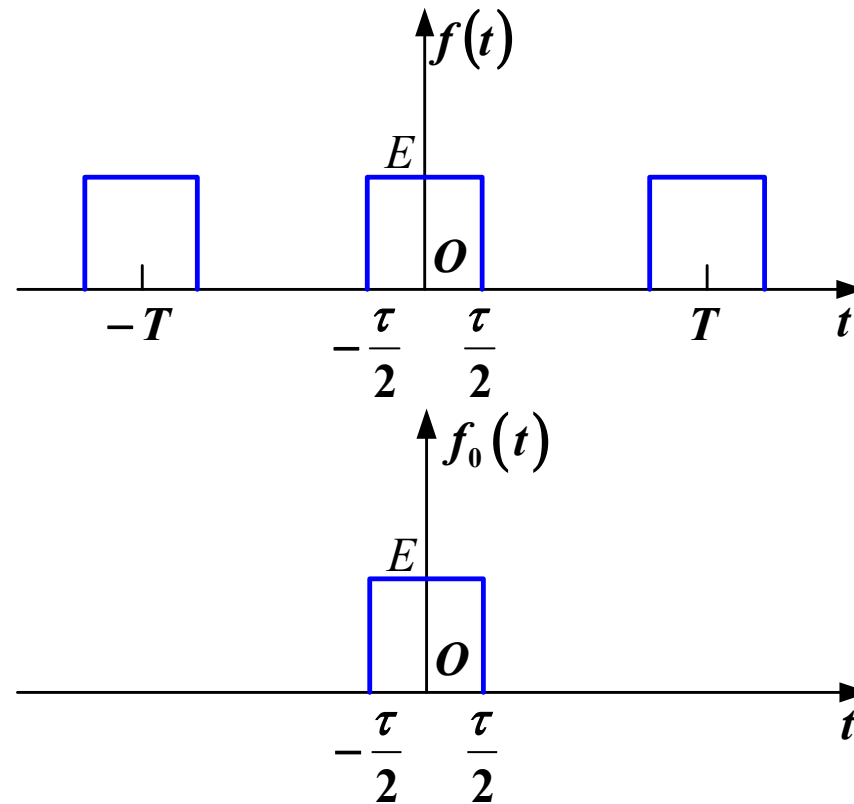
In Polar Form $\mathcal{F}[f(t - t_0)] = |F(\omega)| \cdot e^{j[\varphi(\omega) - \omega t_0]}$

Magnitude spectrum remains unchanged, only affecting the phase spectrum

6. Basic Properties of Fourier Transform

Example 5

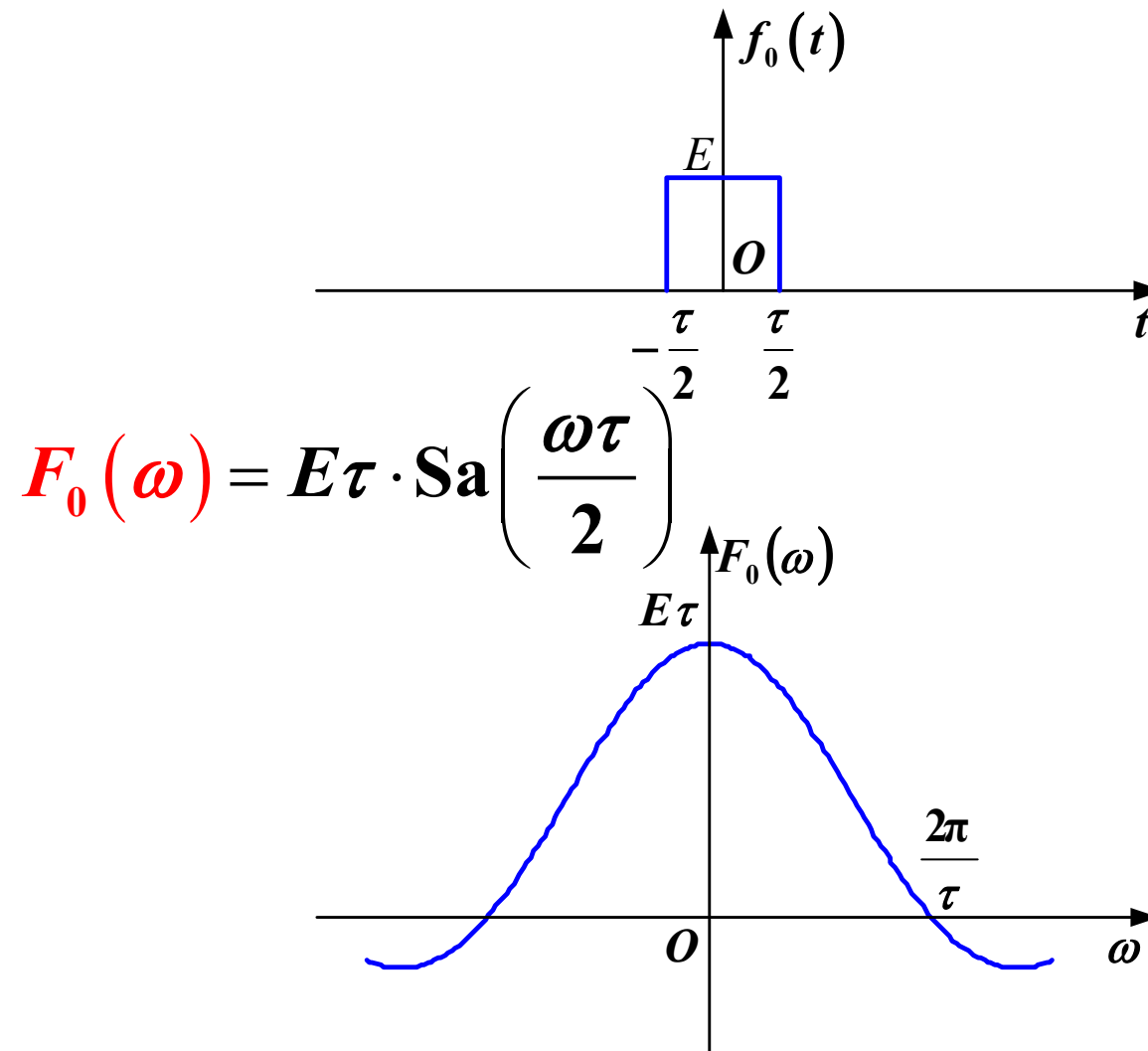
Find the frequency spectrum of the three-pulse signal



$$f(t) = f_0(t) + f_0(t + T) + f_0(t - T)$$

6. Basic Properties of Fourier Transform

Example 5

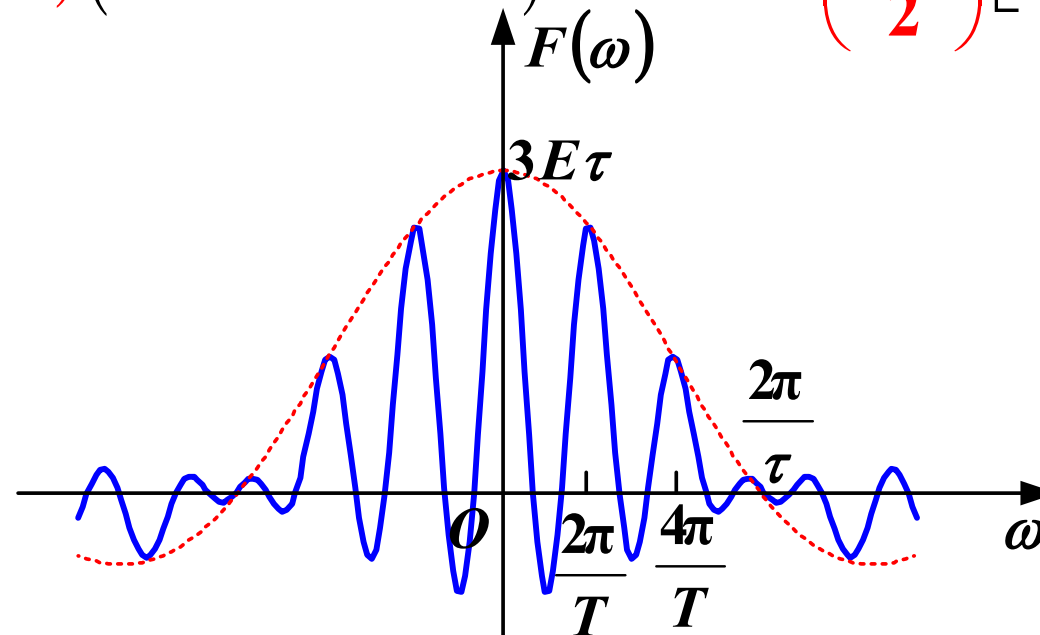


6. Basic Properties of Fourier Transform

Example 5

$$f(t) = f_0(t) + f_0(t + T) + f_0(t - T)$$

$$F(\omega) = F_0(\omega) \left(1 + e^{j\omega T} + e^{-j\omega T} \right) = E\tau \cdot \text{Sa} \left(\frac{\omega\tau}{2} \right) \left[1 + 2\cos(\omega T) \right]$$

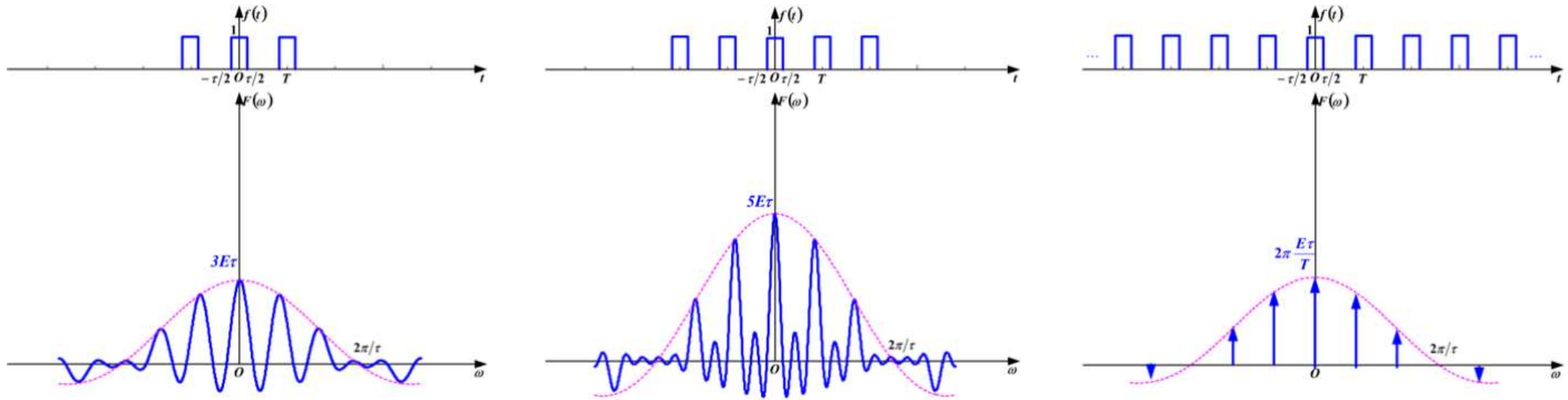


frequency spectrum of
the three-pulse signal

The envelope of the spectrum is determined by the shape of a single pulse

6. Basic Properties of Fourier Transform

Example 5



As the number of pulses increases, the spectral envelope remains unchanged, and the bandwidth stays the same

6. Basic Properties of Fourier Transform

5. Time-Shifting Property

Time-Shifting and Scaling Combined

If $\mathcal{F}[f(t)] = F(\omega)$, then $\mathcal{F}[f(at + b)] = ?$

$$\mathcal{F}[f(at + b)] = \mathcal{F}\left\{f\left[\textcolor{red}{a}\left(t + \frac{\textcolor{blue}{b}}{\textcolor{blue}{a}}\right)\right]\right\} = \frac{\textcolor{red}{1}}{|\textcolor{red}{a}|} F\left(\frac{\textcolor{red}{\omega}}{\textcolor{red}{a}}\right) \cdot \textcolor{blue}{e}^{\textcolor{blue}{j}\omega \frac{b}{a}}$$

Given $\mathcal{F}[f(t)] = F(\omega)$, power spectral density function of $f(2t - 5)$?

$$f(2t) \leftrightarrow \frac{\textcolor{red}{1}}{\textcolor{red}{2}} F\left(\frac{\textcolor{red}{\omega}}{\textcolor{red}{2}}\right)$$

$$f(2t - 5) = f\left[\textcolor{red}{2}\left(t - \frac{\textcolor{blue}{5}}{\textcolor{blue}{2}}\right)\right] \leftrightarrow \frac{\textcolor{red}{1}}{\textcolor{red}{2}} F\left(\frac{\textcolor{red}{\omega}}{\textcolor{red}{2}}\right) \cdot \textcolor{blue}{e}^{-\textcolor{blue}{j}\omega \frac{5}{2}}$$

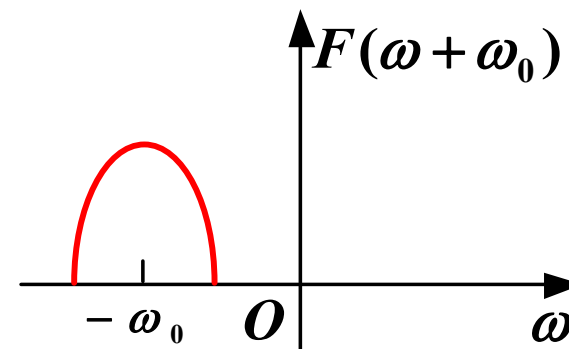
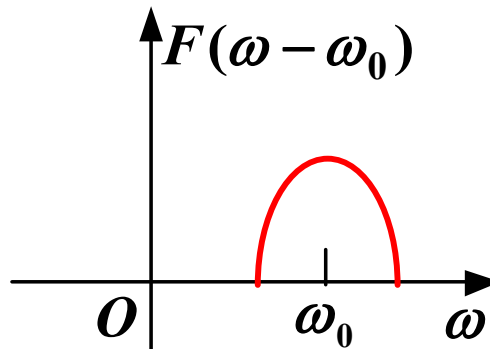
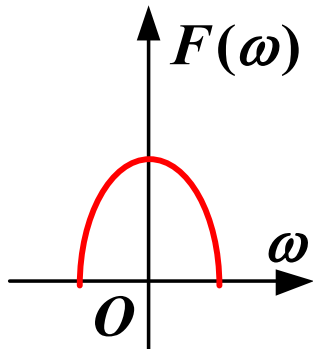
6. Frequency-Shifting Property

Property

$$\text{If } \mathcal{F}[f(t)] = F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$\text{Thus } \begin{cases} \mathcal{F}[f(t) e^{j\omega_0 t}] = F(\omega - \omega_0) \\ \mathcal{F}[f(t) e^{-j\omega_0 t}] = F(\omega + \omega_0) \end{cases}$$

ω_0 is a constant, note: + or -



6. Basic Properties of Fourier Transform

Example 6

Find the frequency spectrum of the signal $f(t) = f_0(t) \cos(\omega_0 t)$

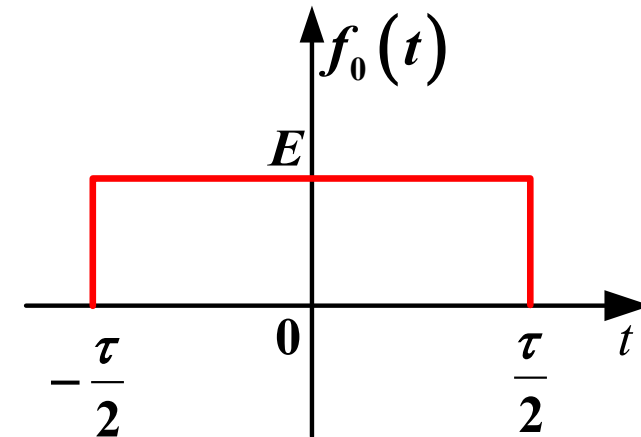
Apply Euler's Formula:

$$f(t) = \frac{1}{2} f_0(t) (e^{j\omega_0 t} + e^{-j\omega_0 t})$$

Use the Frequency-Shifting Property:

$$f(t) \rightarrow F(\omega)$$

$$F(\omega) = \frac{1}{2} F_0(\omega - \omega_0) + \frac{1}{2} F_0(\omega + \omega_0)$$



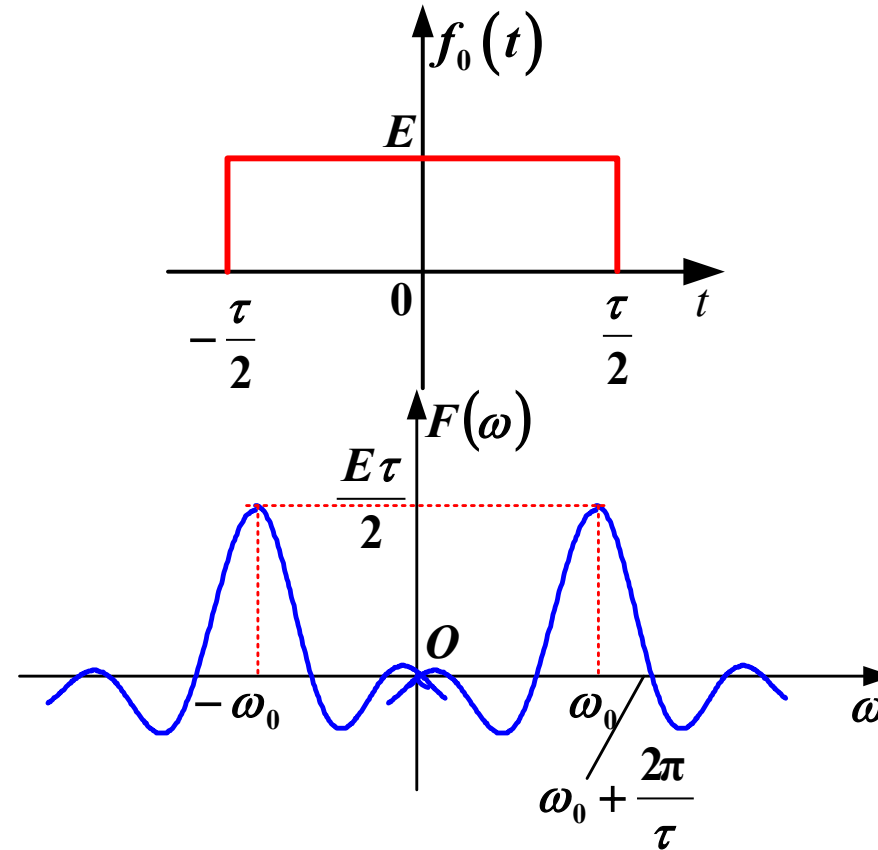
6. Basic Properties of Fourier Transform

Example 6

Given the frequency spectrum $F_0(\omega)$ of the rectangular pulse $f_0(t)$

$$F_0(\omega) = E\tau \cdot \text{Sa}\left(\frac{\omega\tau}{2}\right)$$

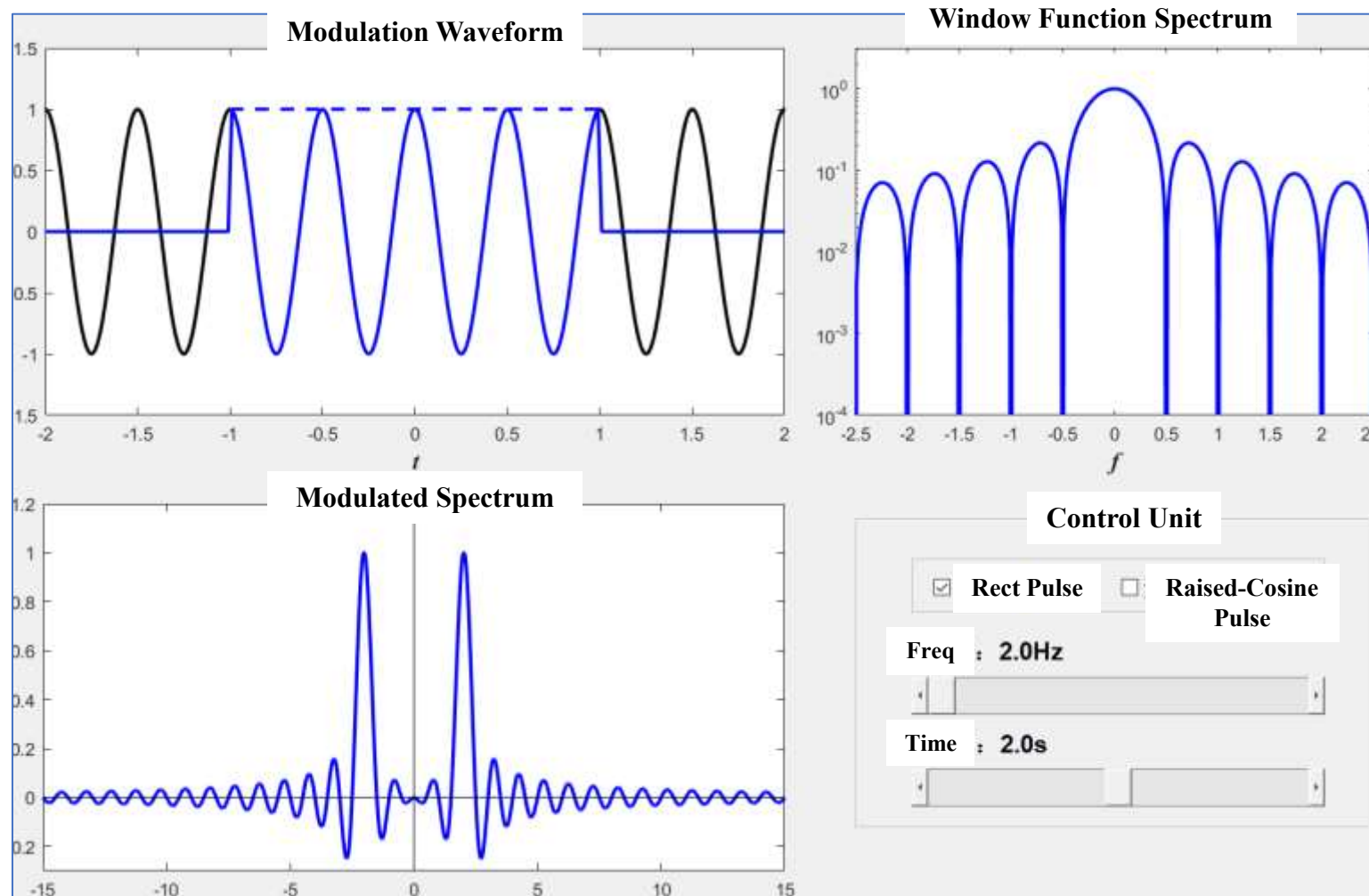
$$F(\omega) = \frac{1}{2} F_0(\omega - \omega_0) + \frac{1}{2} F_0(\omega + \omega_0)$$



The original spectrum $F_0(\omega)$ (a sinc function centered at $\omega_0 = 0$) is split into two copies, shifted to $\omega = \omega_0$ and $\omega = -\omega_0$, scaled by $\frac{1}{2}$ in amplitude

6. Basic Properties of Fourier Transform

Example 6



6. Basic Properties of Fourier Transform

7. Time-Domain Differentiation Property

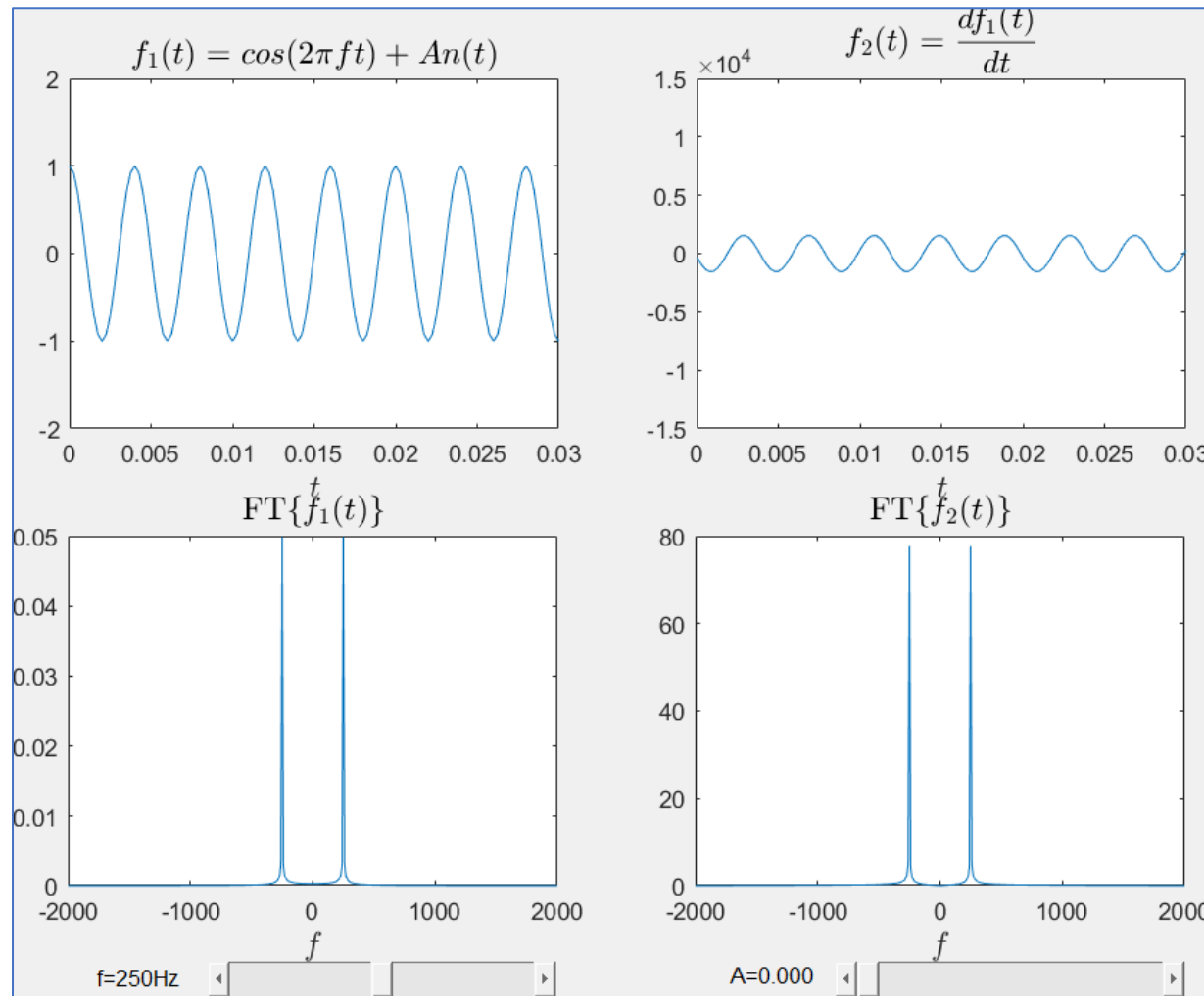
Given $\mathcal{F}[f(t)] = F(\omega)$, then $\mathcal{F}[f'(t)] = j\omega F(\omega)$

$\left\{ \begin{array}{l} \text{The magnitude spectrum is multiplied by } |\omega| \\ \text{phase added, } j = e^{j\frac{\pi}{2}} \end{array} \right.$

Generalization for n-th Derivative: $f^{(n)}(t) \leftrightarrow (j\omega)^n F(\omega)$

6. Basic Properties of Fourier Transform

(1) Demo (Spectrum Obtained via FFT Processing)



6. Basic Properties of Fourier Transform

Example 7

Given $\mathcal{F}[\delta(t)] = 1$, solve $\mathcal{F}[\delta'(t)]$

According to the time-domain differentiation property of the Fourier transform, obtain

$$\mathcal{F}[\delta'(t)] = j\omega \cdot 1$$

8. Frequency-Domain Differentiation Property

$$\text{If } \mathcal{F}[f(t)] = F(\omega), \text{ then } \mathcal{F}[tf(t)] = j \frac{dF(\omega)}{d\omega}$$

$$\text{Or } \mathcal{F}[-jtf(t)] = \frac{dF(\omega)}{d\omega}$$

Generalized Form:

$$\mathcal{F}[(-jt)^n f(t)] = \frac{d^n F(\omega)}{d\omega^n}$$

Or

$$\mathcal{F}[t^n f(t)] = (j)^n \frac{d^n F(\omega)}{d\omega^n}$$

6. Basic Properties of Fourier Transform

Example 9

Given $\mathcal{F}[f(t)] = F(\omega)$, Solve $\mathcal{F}[(t-2)f(t)] = ?$

$$\begin{aligned}\mathcal{F}[(t-2)f(t)] &= \mathcal{F}[tf(t) - 2f(t)] \\ &= j \frac{dF(\omega)}{d(\omega)} - 2F(\omega)\end{aligned}$$

9. Time-Domain Integration Property

If $f(t) \leftrightarrow F(\omega)$, then

$$\int_{-\infty}^t f(\tau) d\tau \leftrightarrow F(\omega) \cdot \left[\frac{1}{j\omega} + \pi\delta(\omega) \right]$$

When $F(0) = 0$, $\int_{-\infty}^t f(\tau) d\tau \leftrightarrow \frac{F(\omega)}{j\omega}$

10. Frequency-Domain Integration Property (for reference)

Given $\mathcal{F}[f(t)] = F(\omega)$,

$$\text{Thus } \mathcal{F}^{-1}\left[\int_{-\infty}^{\omega} F(\omega) d\omega\right] = -\frac{f(t)}{jt} + \pi f(0)\delta(t)$$

11. Time-Domain Convolution Theorem

$$\text{If } \mathcal{F}[f_1(t)] = F_1(\omega), \mathcal{F}[f_2(t)] = F_2(\omega)$$

$$\text{Then } \mathcal{F}[f_1(t) * f_2(t)] = F_1(\omega) \cdot F_2(\omega)$$

Time-domain convolution $\xleftrightarrow{\text{FT}}$ Frequency-domain multiplication

12. Frequency-Domain Convolution Theorem

$$\text{If } \mathcal{F}[f_1(t)] = F_1(\omega), \mathcal{F}[f_2(t)] = F_2(\omega)$$

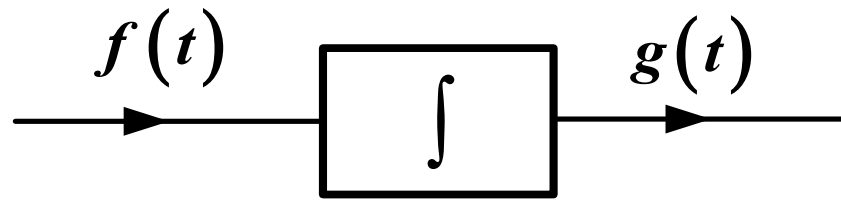
$$\text{Then } \mathcal{F}[f_1(t) \cdot f_2(t)] = \frac{1}{2\pi} F_1(\omega) * F_2(\omega)$$

$$\text{Time-domain multiplication} \xleftrightarrow{\text{FT}} \frac{1}{2\pi} \text{Frequency-domain convolution}$$

13. Applications of the Convolution Theorem

- Using Time-Domain Convolution to Find Spectral Density Functions
- Fourier Transform of the Integral $\int_{-\infty}^t f(\tau) d\tau$
- Fourier Analysis for System (Zero-State) Response (Chapter 4)

13. Applications of the Convolution Theorem



$$g(t) = \int_{-\infty}^t f(\tau) d\tau$$

Unit Impulse Response of an Integrator

$$h(t) = \int_{-\infty}^t \delta(\tau) d\tau = u(t)$$

Response of a Signal Passing Through an Integrator

$$g(t) = f(t) * u(t) = \int_{-\infty}^t f(\tau) d\tau$$

13. Applications of the Convolution Theorem

$$\int_{-\infty}^t f(\tau) d\tau = f(t) * u(t)$$

The time-domain integration property can be obtained through the convolution theorem

$$\begin{aligned}\mathcal{F}\left[\int_{-\infty}^t f(\tau) d\tau\right] &= \mathcal{F}[f(t)] \cdot \mathcal{F}[u(t)] \\ &= F(\omega) \cdot \left[\frac{1}{j\omega} + \pi\delta(\omega)\right]\end{aligned}$$

Summary

1. Linearity
2. Duality
3. Conjugation Symmetry
4. Time-Scaling Property
5. Time-Shifting Property
6. Frequency-Shifting Property
7. Time-Domain Differentiation Property
8. Frequency-Domain Differentiation Property
9. Time-Domain Integration Property
10. Frequency-Domain Integration Property
11. Time-Domain Convolution Theorem
12. Frequency-Domain Convolution Theorem
13. Applications of the Convolution Theorem

Main Content

1. Fourier Transform of Cosine Signal
2. Fourier Transform of Sine Signal
3. Fourier Transform of Impulse Train
4. Fourier Transform of General Periodic Signals

7. Fourier Transform of Periodic Signals

Introduction

Periodic Signal:

The Fourier series F_k of a signal $f(t)$, discrete spectrum

Aperiodic Signal:

The Fourier transform $F(\omega)$ of a signal $f(t)$, continuous spectrum

How to compute the Fourier transform of a **periodic signal**?

What is the relationship with the Fourier series?

$f(t)$,	Periodic	Unified analysis method :
	Aperiodic	Fourier transform analysis method

1. Fourier Transform of Cosine Signal

Euler's formula

$$\cos \omega_0 t = \frac{1}{2} \left(e^{j\omega_0 t} + e^{-j\omega_0 t} \right)$$

Given

$$\mathcal{F}[1] = 2\pi\delta(\omega)$$

Frequency shifting property

$$\mathcal{F}\left[1 \cdot e^{j\omega_0 t}\right] = 2\pi\delta(\omega - \omega_0)$$

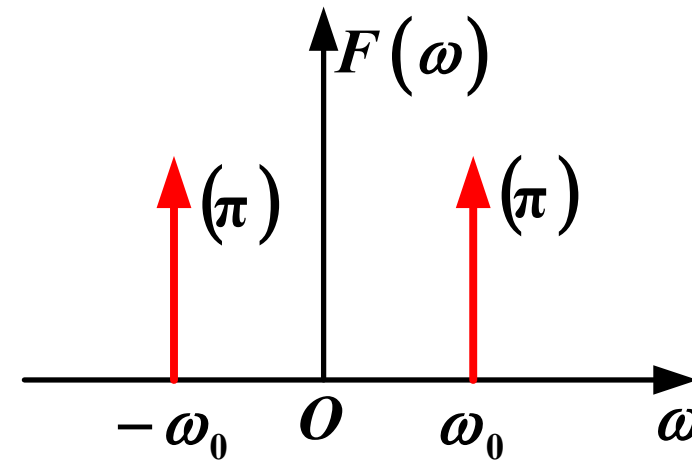
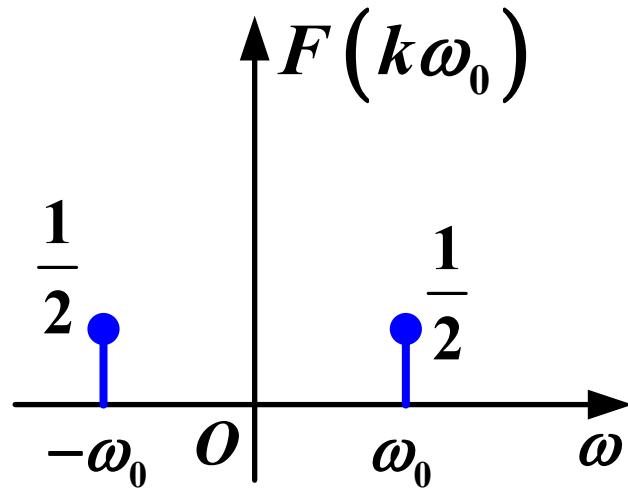
$$\mathcal{F}\left[1 \cdot e^{-j\omega_0 t}\right] = 2\pi\delta(\omega + \omega_0)$$

7. Fourier Transform of Periodic Signals

1. Fourier Transform of Cosine Signal

Fourier transform of cosine signal is

$$\cos \omega_0 t \xleftrightarrow{\text{FT}} \pi \left[\delta(\omega + \omega_0) + \delta(\omega - \omega_0) \right]$$



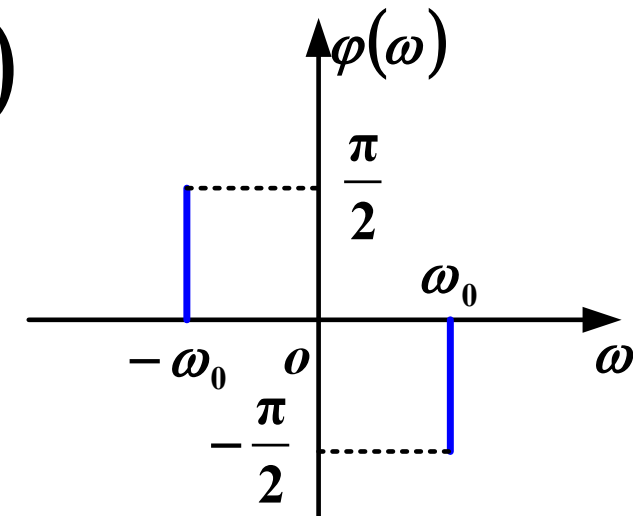
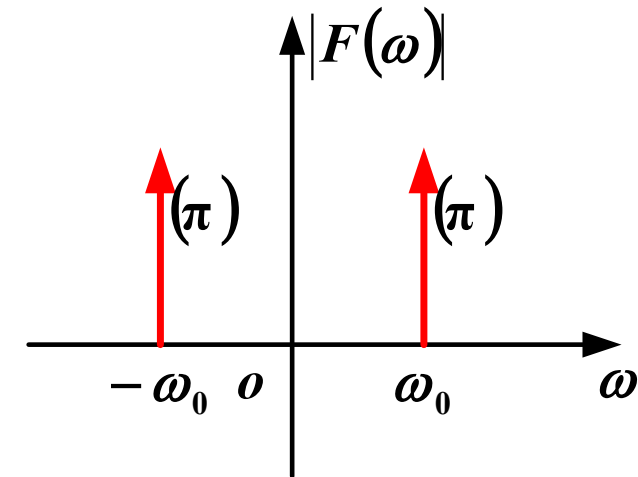
2. Fourier Transform of Sine Signal

$$\sin \omega_0 t = \frac{1}{2j} \left(e^{j\omega_0 t} - e^{-j\omega_0 t} \right)$$

$$\xleftrightarrow{\text{FT}} -j\pi\delta(\omega - \omega_0) + j\pi\delta(\omega + \omega_0)$$

$$= \pi e^{j\left(-\frac{\pi}{2}\right)} \delta(\omega - \omega_0) + \pi e^{j\frac{\pi}{2}} \delta(\omega + \omega_0)$$

- The **amplitude** spectrum of $\sin(\omega_0 t)$ is the same as $\cos(\omega_0 t)$
- The **phase** spectrum is **different**



3. Fourier Transform of Impulse Train

$$\delta_T(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_0) \quad k \text{ is an integer}$$

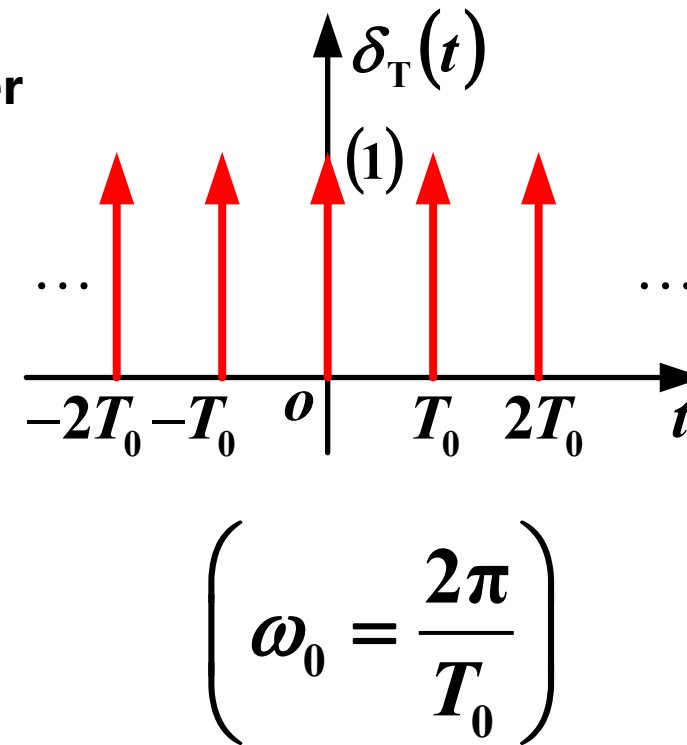
$$\delta_T(t)$$

In the Fourier series, the coefficient is

$$F_k = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \delta(t) e^{-jk\omega_0 t} dt = \frac{1}{T_0}$$

The Fourier series representation

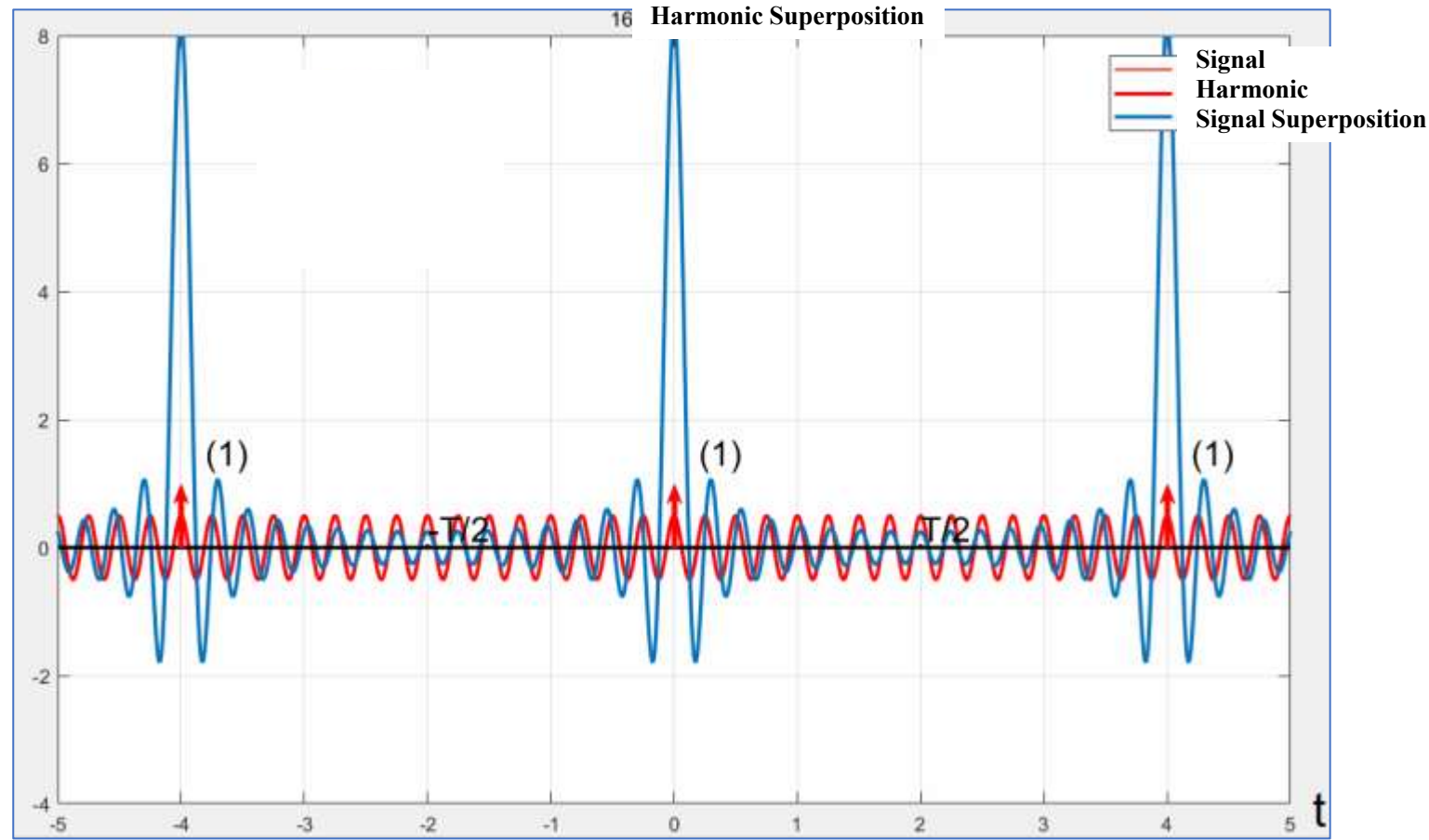
$$\delta_T(t) = \frac{1}{T_0} \sum_{k=-\infty}^{\infty} e^{jk\omega_0 t} = \frac{1}{T_0} + \frac{2}{T_0} \sum_{k=1}^{\infty} \cos(k\omega_0 t)$$



$$e^{j\omega_0 t} + e^{-j\omega_0 t} = 2\cos(\omega_0 t)$$

7. Fourier Transform of Periodic Signals

3. Fourier Transform of Impulse Train



$$\delta_T(t) = \frac{1}{T_0} + \frac{2}{T_0} \sum_{k=1}^{\infty} \cos(k\omega_0 t)$$

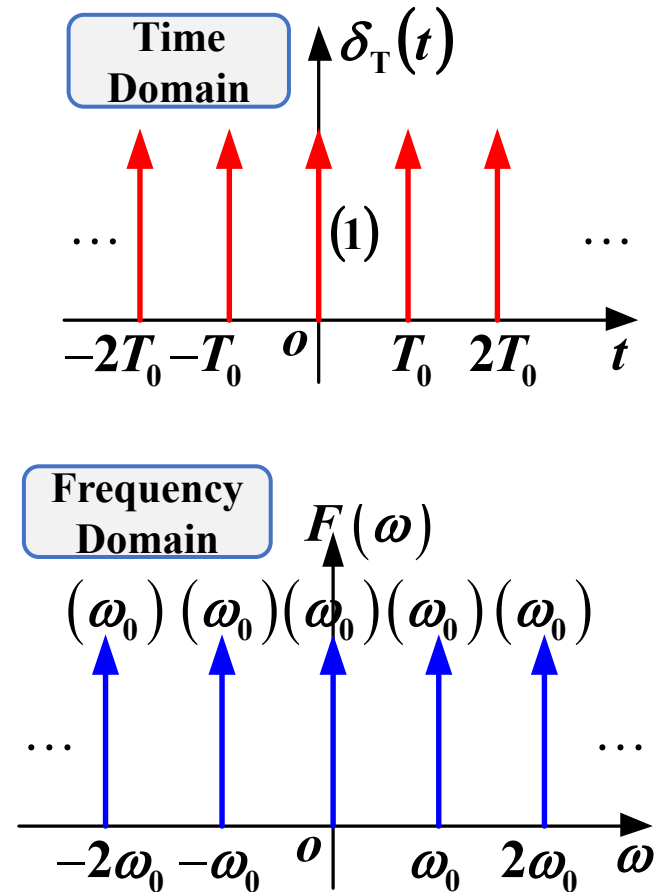
3. Fourier Transform of Impulse Train

$$\delta_T(t) = \frac{1}{T_0} \sum_{k=-\infty}^{\infty} e^{jk\omega_0 t}$$

Given $e^{jk\omega_0 t} \leftrightarrow 2\pi\delta(\omega - k\omega_0)$

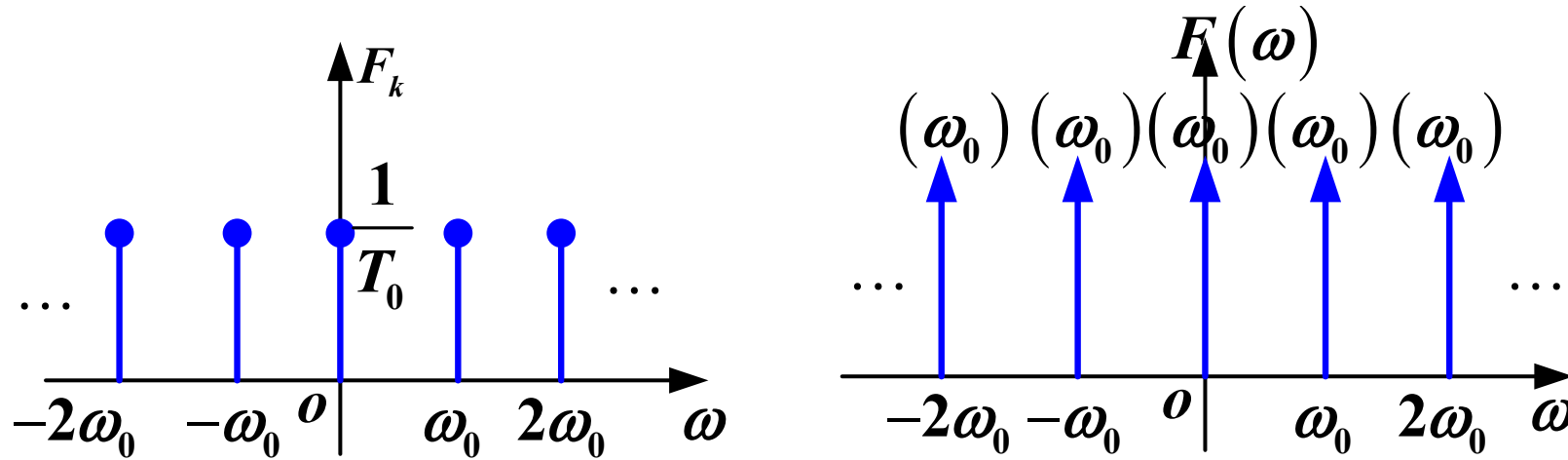
$$\begin{aligned} \therefore F(\omega) &= \frac{1}{T_0} \sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - k\omega_0) \\ &= \omega_0 \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0) \end{aligned}$$

The frequency spectrum of $\delta_T(t)$ remains an **impulse train**, with both strength and spacing of ω_0



7. Fourier Transform of Periodic Signals

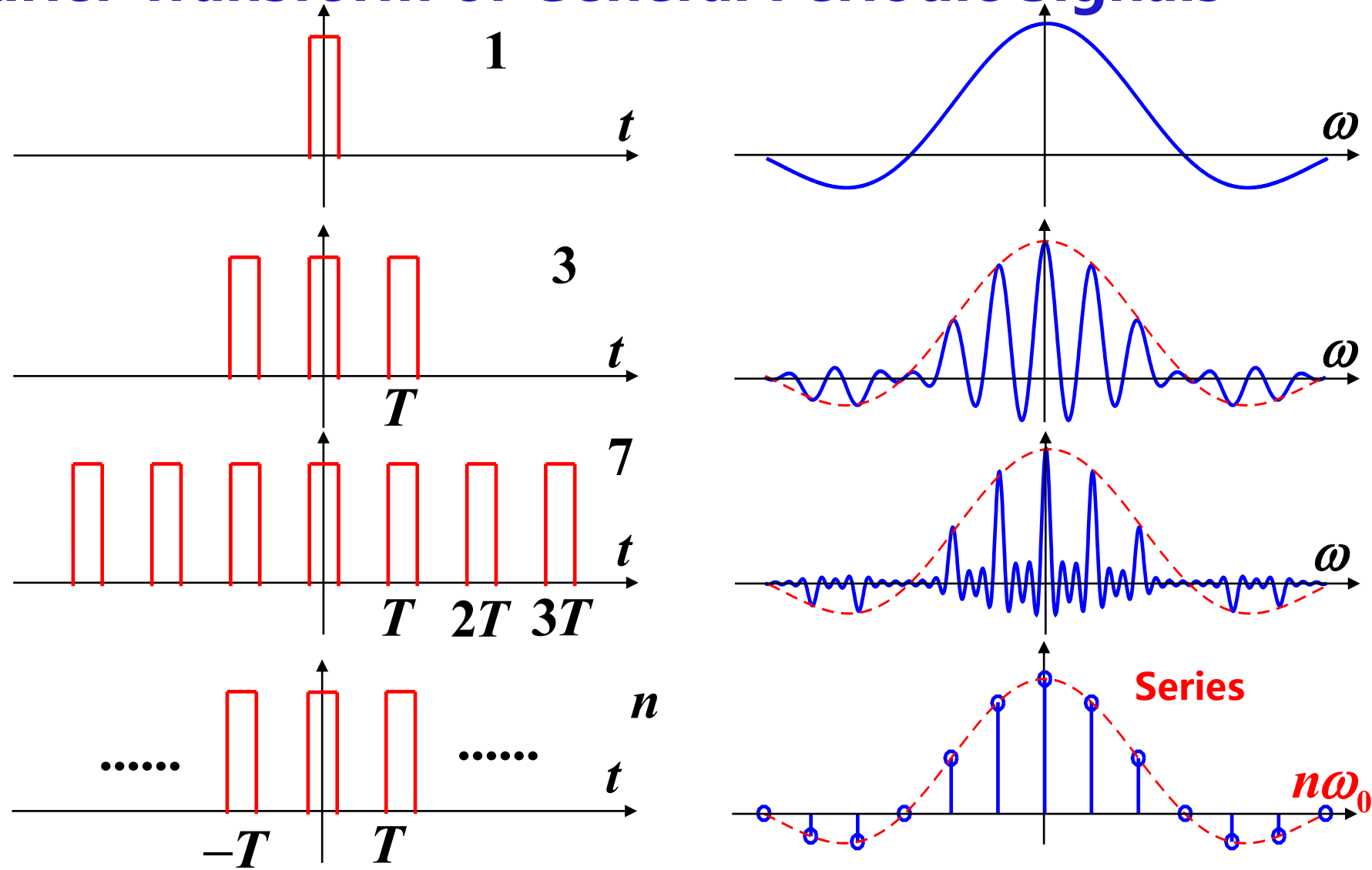
3. Fourier Transform of Impulse Train



**Does not satisfy convergence
Bandwidth is infinitely wide**

7. Fourier Transform of Periodic Signals

4. Fourier Transform of General Periodic Signals



4. Fourier Transform of General Periodic Signals

Periodic signals can be represented in the form of Fourier series

$$f_T(t) = \sum_{k=-\infty}^{\infty} F_k e^{jk\omega_0 t} \quad \left(T_0 = \frac{2\pi}{\omega_0} \right)$$

The Fourier transform of a periodic signal (using the definition)

$$\begin{aligned} F_T(\omega) &= \sum_{k=-\infty}^{\infty} F_k \cdot 2\pi \delta(\omega - k\omega_0) \\ &= 2\pi \sum_{k=-\infty}^{\infty} F_k \cdot \delta(\omega - k\omega_0) \end{aligned}$$

4. Fourier Transform of General Periodic Signals

$$F_T(\omega) = 2\pi \sum_{-\infty}^{\infty} F_k \cdot \delta(\omega - k\omega_0)$$

(1) The spectrum of $f_T(t)$ consists of a series of impulses

Positions: $\omega = k\omega_0$ (harmonic frequencies)

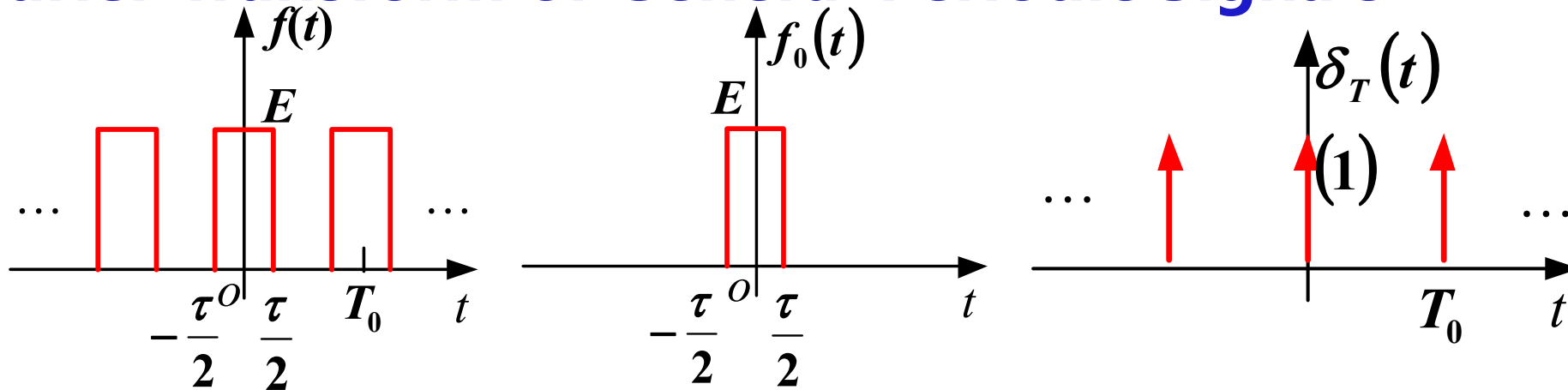
Intensities: $2\pi F_k$, proportional to F_k

(2) The amplitude of spectral lines is not finite because $F(\omega)$ represents spectral density

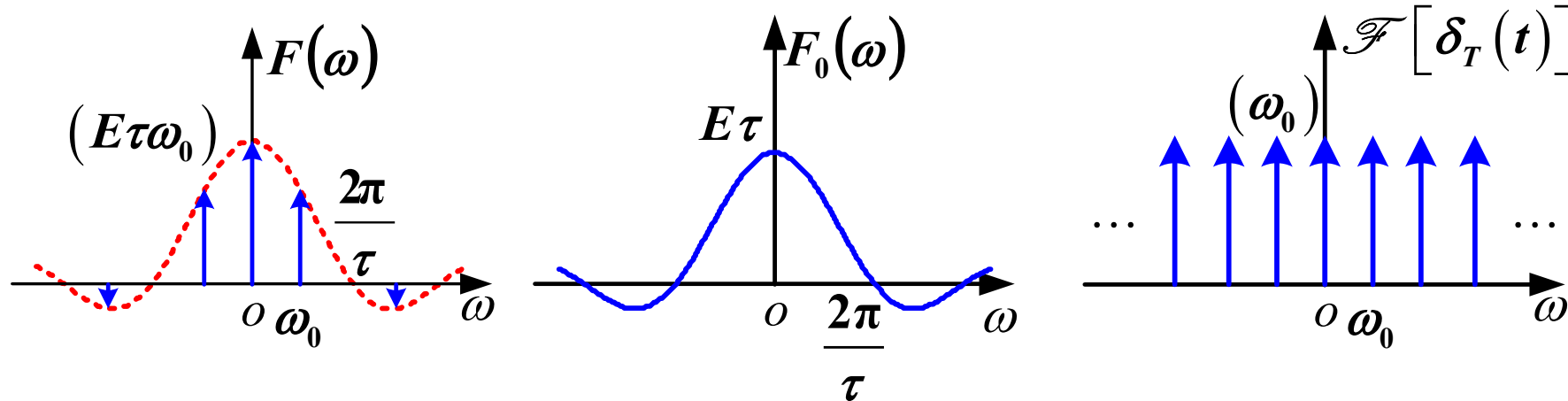
For periodic signals, $F(\omega)$ exists only at discrete points $\omega = k\omega_0$, with an infinitely small frequency range and infinite amplitude

7. Fourier Transform of Periodic Signals

4. Fourier Transform of General Periodic Signals

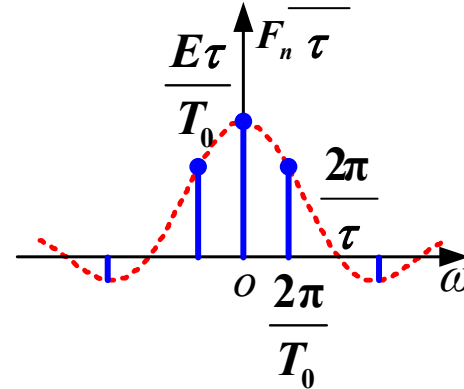
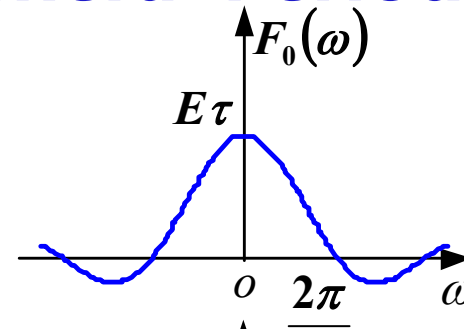
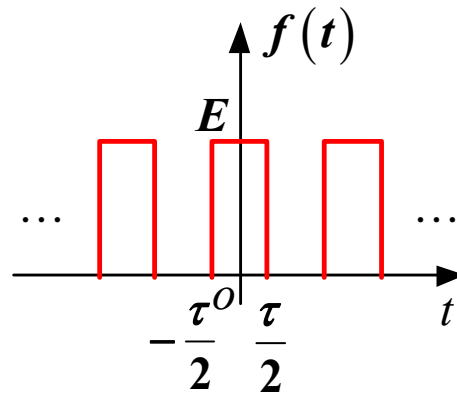
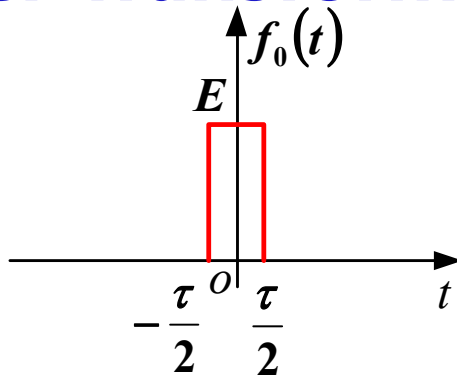


$$F(\omega) = F_0(\omega) \cdot \omega_0 \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0) \quad f(t) = f_0(t) * \delta_T(t)$$

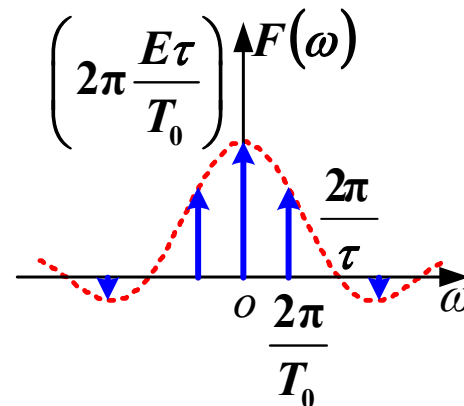


7. Fourier Transform of Periodic Signals

4. Fourier Transform of General Periodic Signals



Fourier series spectrum



Fourier transform spectrum

Summary

1. Fourier Transform of Cosine Signal
2. Fourier Transform of Sine Signal
3. Fourier Transform of Impulse Train
4. Fourier Transform of General Periodic Signals



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Thank you for listening !

